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Scheduling aircraft take-offs and landings on interdependent and heterogeneous runways

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ABSTRACT

This paper presents an optimization method for the aircraft scheduling problem with general runway configurations. Take-offs and landings have to be assigned to a runway and a time while meeting the sequence-dependent separation requirements and minimizing the costs incurred by delays. Some runways can be used only for take-offs, landings, or certain types of aircraft while schedules for interdependent runways have to consider additional diagonal separation constraints.

Our dynamic programming approach solves realistic problem instances to optimality within short computation times. In addition, we propose a rolling planning horizon heuristic for large instances that returns close-to-optimal results.

1. Introduction

Runway systems of major international airports are a bottleneck in the global air traffic network (see, *e.g.*, Balakrishnan and Chandran, 2010; Bennell et al., 2013; Ghoniem et al., 2015). Due to technical restrictions and safety regulations, runway systems only allow a limited number of runway operations (that is, aircraft take-offs and landings) per hour. The total air traffic, however, is growing steadily, and the number of commercial aircraft in use is projected to double within the next two decades (Boeing, 2014). A cost-efficient way to increase the capacity of an existing runway system is to improve its take-off and landing schedules. Thereby, expensive investments in additional runways or airports could be averted or postponed.

The *aircraft scheduling problem* (ASP) can be defined as follows. One set of aircraft is at the gates or on the airfield and is preparing for take-off. Another set of aircraft is approaching the airport by air and is preparing to land. Each aircraft belongs to an aircraft class based on its size and weight, and has a target time for its take-off or landing within a time window. The decision problem at hand is to assign a runway and a take-off or landing time to each aircraft. The resulting runway schedule has to meet separation requirements that are sequence-dependent, as they depend on the operation type, *i.e.*, take-off or landing, and the respective aircraft class of both the preceding and succeeding operation. When an operation is delayed, a delay penalty cost is incurred that depends on the respective operation class, *i.e.*, operation type and aircraft class, and on the length of the delay.

To ensure the necessary separation between runway operations on the same runway, international aviation authorities, such as the FAA (Federal Aviation Administration) and ICAO (International Civil Aviation Organization), define separation requirements for three aircraft classes (small, large, heavy) (see, for example, Table 1a). Note that between two take-offs

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and between two landings, the separation matrix fulfills the triangle inequality. However, when the operation type changes, the triangle inequality is not fulfilled. Consider, for example, the sequence *heavy landing – small take-off – small landing:* The earliest time for the small landing is restricted not by the preceding take-off but by the heavy landing. Thus, when scheduling both take-offs and landings, the separation has to be ensured between all pairs of runway operations. This constraint is referred to as complete separation (Beasley et al., 2000).

We consider general assumptions concerning the airport's runway system that were not previously considered. Many major international airports, such as Frankfurt Airport (the largest German airport) operate runways that are *heterogeneous, i.e.*, not all operations can be performed on all runways, or *interdependent, i.e.*, the operations on one runway also restrict the operations on the other(s). Table 1b shows the separation constraints for parallel runways depending on their spacing. Ashford et al. (2011) give detailed information on the runway specifications required to accommodate certain types of runway operations.

The planning horizon of the ASP is very short: we consider aircraft approaching the airport as soon as they enter the "Extended Terminal Maneuvering Area" (E-TMA) of the airport, approximately 30 to 40 min prior to their target landing time. This area has a radius of up to 40 nautical miles and is controlled by "Terminal Radar Approach Control" (TRACON) (Bennell et al., 2013). We assume to have precise and reliable data on the aircraft on the airport's airfield and in the E-TMA, that is, we have a static and deterministic problem setting. Landing times can be assigned to the approaching aircraft before they reach the final approach path, approximately 20 min before landing. This short planning horizon necessitates a fast solution approach that calculates runway schedules in close to real-time to be of practical use.

Although the ASP has been addressed in many scientific papers, the literature review in Section 2 shows that research on the ASP with both heterogeneous and interdependent runways is scarce. Most papers assume either independent or homogeneous runways, or they restrict themselves to heuristic solution approaches. In this paper, we propose a dynamic programming (DP) approach that derives optimal runway schedules for realistic runway configurations. To solve large problem instances, we also propose a rolling planning horizon (RPH) heuristic that divides a large ASP instance into a number of smaller but connected instances that are then iteratively solved. The main contributions of this paper can be summarized as follows.

- We present a DP-based optimization approach for scheduling runway operations with non-triangular separation times and with interdependent and heterogeneous runways.
- We analyze the efficiency of the proposed approach in a numerical study using realistic data sets and runway systems. For large problem instances, we propose a rolling planning horizon heuristic that yields close-to-optimal results.
- We show the benefit of runway scheduling compared to first-come-first-served (FCFS). Optimized runway schedules are able to handle additional runway operations while still reducing delays.

The remaining part of this paper is organized as follows. In Section 2, we give an overview of related research articles. In Section 3, we describe the model of the ASP with interdependent and heterogeneous runways and formulate a mixed-integer program (MIP). We present the proposed exact solution approach in Section 4 and the heuristic approach in Section 5. We

Table 1
Separation requirements (in seconds).

(a) Separation for operations on the same runway

			Trailing aircraft					
			Landing			Take-off		
			Small	Large	Heavy	Small	Large	Heavy
Leading aircraft	Landing	Small	82	69	60	75	75	75
		Large	131	69	60	75	75	75
		Heavy	196	157	96	75	75	75
	Take-off	Small	60	60	60	60	60	60
		Large	60	60	60	60	60	60
		Heavy	60	60	60	120	120	90

Source: *e.g.*, Balakrishnan and Chandran (2010) and Farhadi et al. (2014)

(b) Separation for operations on parallel runways

Runway spacing	Take-off → Take-off	Take-off → Landing	Landing → Take-off	Landing → Landing
Up to 2500 ft (up to 760 m)	As on single runway	As on single runway	Independent (no separation)	As on single runway
2500–4300 ft (760–1310 m)	Independent	Independent	Independent	40 s
More than 4300 ft (more than 1310 m)	Independent	Independent	Independent	Independent

Source: De Neufville and Odoni (2013)

conduct a numerical study in Section 6. In Section 7, we outline how costs for taxiing aircraft from their gate to the runway and vice versa can be integrated in our model. Section 8 concludes the paper and outlines future research opportunities.

2. Related literature

The ASP is a well-established optimization problem that has been subject to intensive study. The first solution approaches for a basic version of this problem with a single runway and without positive target times were presented by Dear (1976) and Psaraftis (1980). If separation requirements or delay penalty costs are assumed to be aircraft-dependent instead of operation class-dependent, the ASP can be shown to be NP-hard (Bianco et al., 1999). In this paper, however, we assume that both separation requirements and delay cost functions are operation class-dependent. Briskorn and Stolletz (2014) prove that, under these assumptions, the ASP is polynomial in the number of runway operations but exponential in the number of runways and operation classes by defining an exact DP with a polynomial state space. Bennell et al. (2013) provide an extensive literature overview on airport runway operations, featuring articles up to 2009. Section 2.1 reviews models for the ASP that consider heterogeneous or interdependent runways. Related solution approaches are discussed in Section 2.2.

2.1. Overview of models

The number of papers on the ASP that consider heterogeneous or interdependent runways, is limited. *Heterogeneous runways* are considered by Hansen (2004), Pinol and Beasley (2006), Salehipour et al. (2009), and Liu (2011). All papers assume runway-dependent earliest operation times. In Salehipour et al. (2009) and Liu (2011), this is the only heterogeneous feature of the runway systems. Hansen (2004), in addition, assumes that not all operations can be performed on all runways. Pinol and Beasley (2006) model runway-dependent target times and separation constraints in a MIP but restrict their numerical study to runway-independent problem instances. All four papers provide only heuristic results to their problem instances.

Interdependent runways are considered in most cited MIP formulation by Beasley et al. (2000) through an additional matrix of separation times for all pairs of operations on different runways. By this definition, interdependence is assumed to be the same for all pairs of runways. Pinol and Beasley (2006), Bencheikh et al. (2011), Xiangwei et al. (2011), Salehipour et al. (2013), Faye (2015), Sabar and Kendall (2015) adopt this definition, but none of these papers consider interdependent runways in their numerical studies.

Bianco et al. (2006) take approach paths and exit routes into consideration by modeling the ASP as a no-wait job-shop problem. They consider airports with two parallel runways (Milan–Malpensa and Rome-Fiumicino) and assume an aircraft class-independent minimum diagonal separation, as in Table 1b. This approach is generalized by Samà et al. (2013, 2014a,b, 2015), and D'Ariano et al. (2015). These papers use alternative graphs to model holding circles, approach paths, and exit routes, and their objective is to minimize the maximum (consecutive) delay on two-runway systems with interdependent runways. In addition, Samà et al. (2014a,b, 2015) consider integrated aircraft routing decisions in the TMA, D'Ariano et al. (2015) consider collaborative decision-making with en-route air traffic controllers, and Samà et al. (2015) consider a combined objective of minimizing delays and travel times in the TMA.

Table 2 gives an overview of the discussed papers. Column 2 shows the runway configuration under consideration, *i.e.*, the number of runways and whether they are interdependent or heterogeneous. Column 3 shows the objective function(s). The most common objective functions maximize the runway throughput, *i.e.*, minimize the makespan, minimize total (weighted) earliness and tardiness (E/T) or minimize the total (weighted) delays. Lee and Balakrishnan (2008) show numerically that delay minimization is a favorable objective as it usually entails a high runway throughput, while throughput maximization may result in excessive delays. Columns 4 and 5 list the applied solution approaches and data used in the numerical studies, respectively. The last column states additional constraints or distinguishing features of the respective paper. None of the papers provide an efficient optimization approach for the ASP considering interdependent as well as heterogeneous runways.

2.2. Solution approaches

This section presents the solution approaches from the papers discussed in Section 2.1. In addition, we review recent papers (from 2010 to 2015) on the less general problem with single or independent runways that were not covered in the survey of Bennell et al. (2013). An overview of these papers is given in Table 3. We present heuristic approaches in Section 2.2.1, approximate approaches in Section 2.2.2, and exact approaches in Section 2.2.3.

2.2.1. Heuristic solution approaches

The majority of papers on the ASP focus on developing heuristic solution approaches. These approaches comprise column generation, dispatch rules, decomposition, and a large variety of metaheuristics.

Column generation: Ghoniem and Farhadi (2015) present the first efficient column generation approach with stabilization mechanisms for the ASP with multiple, independent runways. The approach is not optimizing as it does not use Branch-&-Price to ensure integer optimal solutions.

Dispatch rules: Hancerliogullari et al. (2013) propose different dispatch rule-based and metaheuristic solution approaches. Ghoniem et al. (2014) present a rule-based approach ("optimized FCFS") and a MIP-based approach ("threshold-based suboptimized heuristic"). They evaluate their solution quality using practice data from Doha International Airport. Farhadi et al. (2014) use the same data sets to evaluate different dispatch rule-based heuristics, runway configurations, and separation standards. In addition to an exact Branch-&-Bound approach, D'Ariano et al. (2015) solve their aircraft scheduling and routing problem with different dispatch rules and greedy heuristics.

Decomposition: A common approach to decompose the ASP into smaller, computationally tractable problems is to consider only a limited time window of the planning horizon at a time and derive schedules for these smaller time windows iteratively. This approach is referred to as rolling planning horizon (RPH), receding horizon, or sliding window and has been analyzed by Xiangwei et al. (2011), Samà et al. (2013, 2014a,b), and Furini et al. (2015). RPH can be used to apply the (static) ASP in a dynamic environment and for longer planning horizons. Besides decomposition by time, the ASP with integrated aircraft routing in the TMA can also be decomposed into routing and scheduling decisions (Samà et al., 2014a,b).

Metaheuristic approaches: Hansen (2004) presents a genetic algorithm (GA) for the ASP with heterogeneous runways. Pinol and Beasley (2006) present two population-based heuristic solution approaches for interdependent and heterogeneous runways. Bianco et al. (2006) take approach paths and exit routes into consideration. They define the ASP as a no-wait job-shop problem and provide a heuristic local search algorithm for a two-runway airport with interdependent runways. Salehipour et al. (2009) present simulated annealing (SA) and variable neighborhood search (VNS) methods, and Liu (2011) presents a GA. Both papers consider heterogeneous runways. Bencheikh et al. (2011) and Ma et al. (2014) describe ant colony optimization (ACO) approaches for the ASP. Bencheikh et al. (2011) consider interdependent runways, while Ma et al. (2014) only consider a single runway. Salehipour et al. (2013) present SA and VNS approaches for interdependent runways. Vadlamani and Hosseini (2014) propose an adaptive large neighborhood search (ALNS) algorithm for a single-runway setting. Sabar and Kendall (2015) present an iterated local search (ILS) heuristic with different perturbation operators for interdependent runways.

2.2.2. Approximation approaches

Approximation approaches seek to reduce the computational complexity of the ASP, accepting a bounded loss of solution quality. One approach is to discretize the time axis: Heidt et al. (2014) and Faye (2015) present time-indexed MIP formulations and Branch-&-Bound approaches to solve these problems efficiently. Balakrishnan and Chandran (2010) present a DP approach for the time-indexed ASP assuming a single runway and constrained position shifting (CPS). In CPS, for each runway operation, they allow only a limited deviation from the FCFS sequence. Ma et al. (2014) and Faye (2015) propose to approximate the separation matrix by a rank 2 matrix, *i.e.*, by a matrix that can be expressed as the sum of 2 vectors. In this manner, the ASP becomes sequence-independent and thus easier to solve, but a significant loss of precision is incurred. Ma et al. (2014) consider a single-runway setting while Faye (2015) considers interdependent runways as in Beasley et al. (2000).

2.2.3. Optimizing solution approaches

Three classes of optimizing approaches for the ASP are described in the literature: mixed-integer programming, dynamic programming, and Branch-&-Price.

Mixed-integer programming models: A common optimization approach for the ASP is to formulate a MIP and solve it using a standard solver, such as CPLEX. The most cited MIP formulation by Beasley et al. (2000) considers complete separation and interdependent runways. Recently, several authors proposed reformulations or valid inequalities (VI) to tighten this formulation (Briskorn and Stolletz, 2014; Farhadi et al., 2014; Ghoniem et al., 2014; Ghoniem and Farhadi, 2015). They generally outperform the MIP by Beasley et al. (2000) but still exhibit prohibitive computation times for all except small problem instances, and they do not consider interdependent runways. Samà et al. (2015) and D'Ariano et al. (2015) provide Branch-and-Bound approaches for the aircraft scheduling and routing problem.

Dynamic programming approaches: DP approaches for the ASP are presented by, *e.g.*, Balakrishnan and Chandran (2010), Harikiopoulo and Neogi (2011), Furini et al. (2014), Lieder et al. (2015). These approaches exhibit fast computation times but have additional assumptions that simplify the underlying problem. Except for that of Lieder et al. (2015), all approaches assume a single-runway problem. Balakrishnan and Chandran (2010) assume discrete time and CPS. Furini et al. (2014) generalize this approach to continuous time. Harikiopoulo and Neogi (2011) present an efficient DP algorithm for makespan minimization. Lieder et al. (2015) propose a DP-based optimization algorithm with fast computation times for multiple runways. This approach only ensures successive separation and assumes identical and independent runways.

Branch-&-Price: Ghoniem et al. (2015) propose a Branch-&-Price approach that outperforms standard solvers. As they use single-runway problems as subproblems to generate columns, they emphasize that their approach cannot be applied to interdependent runways.

To solve the ASP as outlined in Section 1, we generalize the DP approach by Lieder et al. (2015). Among the existing exact approaches, DP provides better opportunities for generalization than Branch-&-Price as it is more adaptive with respect to adding constraints and modeling general runway configurations.

Table 2
Overview of related literature with heterogeneous or interdependent runways.

Source	No. of runways & configuration	Objective function(s)	Solution approach(es)	Numerical data	Remarks
Beasley et al. (2000)	$\geqslant 1$, interdependent	min $\sum$ w. E/T	MIP, time-indexed MIP, Branch-&-Bound	OR-Library (Beasley, 1990)	Most-cited MIP
Hansen (2004)	$\geqslant 1$, heterogeneous	min max delay	GA	Practice (Dallas Fort Worth, 12–20 aircraft)	
Bianco et al. (2006)	1–2, interdependent	min makespan, min avg. delay	Job shop model, local search heuristic	practice (Milan, Rome)	Modeling of approach paths, no exact solution approach
Pinol and Beasley (2006)	$\geqslant 1$, interdep., heterogeneous	min $\sum$ w. E/T	MIP, 2 population-based heuristics	Beasley (1990)	
Salehipour et al. (2009)	$\geqslant 1$, heterogeneous	min $\sum$ delay	VNS	Hansen (2004)	
Bencheikh et al. (2011)	$\geqslant 1$, interdependent	min $\sum$ w. E/T	ACO	Beasley (1990)	Assumption of symmetric separation matrix
Liu (2011)	$\geqslant 1$, heterogeneous	min $\sum$ delay (squared)	MIP, local search	Beasley (1990), Hansen (2004)	Runway-dependent time windows, only heuristic results
Xiangwei et al. (2011)	$\geqslant 1$, interdependent	min $\sum$ w. E/T	RPH	Beasley (1990),	
Salehipour et al. (2013)	$\geqslant 1$, interdependent	min $\sum$ w. E/T	MIP, SA, VNS	Beasley (1990)	
Samà et al. (2013)	2, interdependent	min max delay	B&B, RPH	Practice (Milan, Rome)	Modeling of approach paths, generalization of Bianco et al. (2006)
Samà et al. (2014a) Samà et al. (2014b)	2, interdependent	min max delay	B&B, RPH, tabu search, decomposition	Practice (Milan)	Modeling of approach paths and aircraft routing decisions
D'Ariano et al. (2015)	2, interdependent	min max delay	B&B, dispatch rules, greedy heuristics	Practice (Milan, Rome)	modeling of approach paths, collab. with en-route traffic control
Faye (2015)	$\geqslant 1$, interdependent	min $\sum$ w. E/T	Time-indexed MIP, MIP-based approximation	Beasley (1990)	Time-indexed MIP fast for small instances
Sabar and Kendall (2015)	$\geqslant 1$, interdependent	min $\sum$ w. E/T	ILS	Beasley (1990)	
Samà et al. (2015)	2, interdependent	min max delay, & $\sum$ travel times	MIP, B&B	Practice (Rome)	Approach paths, routing decisions, multi-criteria objectives
This paper	$\geqslant 1$, interdep., heterogeneous	min $\sum$ w. delay	MIP, DP	Generated (30–62 aircraft)	Exact approach, most general runway settings

Table 3
Overview of related literature with single or independent runways (2010–2015).

Source	No. of runways & configuration	Objective function(s)	Solution approach(es)	Numerical data	Remarks
Balakrishnan and Chandran (2010)	1	min makespan, min $\sum$ delay, minmax delay	Discrete-time DP with CPS	Generated (10–50 ac.)	Discontinuous time windows possible
Harikiopoulo and Neogi (2011)	1	min makespan	DP	Generated (10–110 ac.)	Only successive separation
Hancerliogullari et al. (2013)	$\geqslant 1$, independent	min $\sum$ w. delay	MIP, dispatch rules, SA	Generated (15–25 ac.)	
Briskorn and Stolletz (2014)	$\geqslant 1$, independent	min $\sum$ w. E/T	MIP with aircraft classes, DP (description)	Bianco et al. (1999) (MIP only)	Proof of polynomiality (w. classes), no numerical study of DP
Farhadi et al. (2014)	$\geqslant 1$, independent	min $\sum$ w. delay	MIP, dispatch rules, MIP-based heuristic	Practice (Doha)	
Furini et al. (2014)	1	min $\sum$ w. E/T	MIP with CPS, continuous-time DP with CPS	Practice (Milan), Beasley (1990)	Generalization of Balakrishnan and Chandran (2010)
Ghoniem et al. (2014)	1	min makespan	MIP with VI, dispatch rules	Practice (Doha), Generated (20–40 ac.)	
Heidt et al. (2014)	1	min $\sum$ w. E/T (squared)	Time-indexed MIP with different period sizes	Generated (50 ac.)	Robustness considerations, open time windows
Ma et al. (2014)	1	min makespan	MIP, approximation, ACO	Generated (20–200 ac.)	Only successive separation, no optimal results reported
Vadlamani and Hosseini (2014)	1	min $\sum$ w. E/T	ALNS	Beasley (1990)	
Furini et al. (2015)	1	min $\sum$ w. delay	Position-based MIP, RPH	Practice (Milan)	Fast results for small CPS values
Ghoniem and Farhadi (2015)	$\geqslant 1$, independent	min $\sum$ w. E/T, min $\sum$ w. comp.	MIP with VI, Column generation	Beasley (1990), generated (15–50 ac.)	First efficient column generation approach
Ghoniem et al. (2015)	$\geqslant 1$, independent	min $\sum$ w. delay	Column generation with Branch-&-Price	Generated (15–50 ac.)	Efficient exact method for independent runways
Lieder et al. (2015)	$\geqslant 1$, independent	min $\sum$ w. delay	MIP, DP	Bianco et al. (1999), generated (50–100 ac.)	Only successive separation

3. Model description

We consider a given set of aircraft, A. Each aircraft $a \in A$ belongs to one class out of a set of operation classes, W. Each aircraft's operation class is defined by its size and by whether the aircraft is to perform a take-off or landing. We denote aircraft a's operation class by $w(a)$. Each aircraft has a target operation time, T_a, and a latest possible operation time, L_a. The scheduled operation time of aircraft a is denoted by C_a and has to be within a's time window, *i.e.,* $T_a \leqslant C_a \leqslant L_a$. If the operation of aircraft a is scheduled later than its target time, a delay cost is incurred according to a cost function, $c(w(a), C_a - T_a)$, that depends on the respective aircraft's operation class and delay. We assume that the aircraft in set A are ordered by target time, T_a.

The airport under consideration has a set of runways, R. The operation of each aircraft $a \in A$ must be assigned to exactly one runway. It must be ensured that the runway operation is allowed on the assigned runway. A runway operation can be defined as a tuple, (a, r, C_a), with runway r being able to handle operations of class $w(a)$ and with $T_a \leqslant C_a \leqslant L_a$.

Between each pair of runway operations, a minimum separation time must hold. As we assume runway systems that can contain both independent and interdependent runways, this separation time depends on both the class of the preceding and succeeding operation and the runways that both operations are assigned to. $Sep^{r,r'}_{w(a),w(a')}$ denotes the minimum separation time between the operation time of aircraft a and the operation time of aircraft a' if aircraft a is assigned to runway r and aircraft a' is assigned to runway r'. If r and r' are independent, the respective separation time is zero.

A *runway schedule* is a feasible solution to the ASP. It is defined by a set of runway operations where.

1. there is exactly one runway operation (a, r, C_a) for each aircraft $a \in A$, and
2. between all pairs of runway operations, (a, r, C_a) and $(a', r', C_{a'})$ with $C_a \leqslant C'_a$, the minimum separation time $Sep^{r,r'}_{w(a),w(a')}$ is upheld.

An *optimal solution* to the ASP is a runway schedule with minimal total delay penalty cost. We assume that the delay penalty cost functions, $c(\cdot)$, are piecewise linear and convex. This allows for, *e.g.,* penalizing large delays with a higher cost increase than small delays or making small delays "free," *i.e.,* without any penalty costs. We also assume ordered time windows, that is, we assume that the latest possible operation times of all pairs (a, a') of aircraft are in the same order as their target times, *i.e.,* $T_a \leqslant T_{a'} \iff L_a \leqslant L_{a'}$. Under these assumptions, all feasible ASP instances have an optimal solution, in which the aircraft of the same aircraft class are scheduled FCFS, *i.e.,* for $w(a) = w(a') \wedge T_a < T_{a'},\ C_a \leqslant C_{a'}$ holds for all $(a, a' \in A;\ a \neq a')$ (Briskorn and Stolletz, 2014; Ghoniem et al., 2014). There are no sequencing restrictions for pairs of operations from different classes.

The following mixed-integer optimization model is a generalization of the model formulation by Beasley et al. (2000). The model therein assumes that runways are either independent or that the diagonal separation times are the same for all pairs of runways. We define more general separation times, $Sep^{r,r'}_{w(a),w(a')}$, that depend on the preceding operation, the succeeding operation, and the respective pair of runways.

In addition to the continuous decision variable, C_a, that defines the operation start time of each aircraft $a \in A$, we have a binary variable $\delta_{a,a'}$ that defines the overall sequence of all operations, a binary variable y^r_a, that defines the runway assignment of each operation, and an auxiliary binary variable $z^{r,r'}_{a,a'}$ that defines the runway assignments of all pairs of operations. The sets, parameters, and decision variables are summarized in Table 4.

$$\text{Minimize} \quad F = \sum_{a \in A} c(w(a), C_a - T_a) \tag{1}$$

subject to the constraints:

$$T_a \leqslant C_a \leqslant L_a \qquad \forall a \in A \tag{2}$$

$$\delta_{a,a'} + \delta_{a',a} = 1 \qquad \forall a, a' \in A;\ a \neq a' \tag{3}$$

$$\delta_{a,a'} = 1 \qquad \forall a, a' \in A;\ a \neq a';\ w(a) = w(a');\ T_a < T_{a'} \tag{4}$$

$$C_a + Sep^{r,r'}_{w(a),w(a')} \cdot z^{r,r'}_{a,a'} \leqslant C_{a'} + M\delta_{a',a} \qquad \forall a, a' \in A;\ a \neq a';\ \forall r, r' \in R;\ Sep^{r,r'}_{w(a),w(a')} > 0 \tag{5}$$

$$\textstyle\sum_{r \in R} y^r_a = 1 \qquad \forall a \in A \tag{6}$$

$$y^r_a \leqslant P^r_{w(a)} \qquad \forall a \in A;\ \forall r \in R \tag{7}$$

$$z^{r,r'}_{a,a'} \geqslant y^r_a + y^{r'}_{a'} - 1 \qquad \forall a, a' \in A;\ a \neq a';\ \forall r, r' \in R \tag{8}$$

The objective function (Eq. 1) minimizes the total penalty cost for delayed operations. Constraint (2) ensures that each aircraft is scheduled within its time window. Constraint (3) defines a sequence over all pairs of runway operations. The order of same-class operations is fixed due to the "FCFS within operation classes" property by Eq. (4). Employing a sufficiently large number, M, constraint (5) ensures the separation requirements for all pairs of operations on all pairs of runways, *i.e.,* complete separation and diagonal separation hold. A sufficiently large M for a pair $((a, r, C_a), (a', r', C_{a'}))$ of operations is, *e.g.,*

Table 4
Sets, parameters, and variables of the MIP formulation.

Sets:		
A	Set of aircraft	
W	Set of aircraft classes	
R	Set of runways	
Parameters:		
$w(a)$	Class of aircraft a; $w(a) \in W$	
T_a	Target take-off/landing time of aircraft a	
L_a	End of time window of aircraft a	
$Sep^{r,r'}_{w(a),w(a')}$	Aircraft class-dependent minimum separation time between aircraft a (on runway r) and aircraft a' (on runway r')	
$P^r_{w(a)}$	$= \begin{cases} 1 & \text{if runway } r \text{ can be used by aircraft class } w(a) \\ 0 & \text{otherwise} \end{cases}$	
Decision variables:		
C_a	Assigned operation time of aircraft $a \in A$	
$\delta_{a,a'}$	$= \begin{cases} 1 & \text{if aircraft } a \text{ is scheduled before aircraft } a' \\ 0 & \text{otherwise} \end{cases}$	$\forall a \neq a' \in A$
y^r_a	$= \begin{cases} 1 & \text{if aircraft } a \text{ is assigned to runway } r \\ 0 & \text{otherwise} \end{cases}$	$\forall a \in A,\ r \in R$
$z^{r,r'}_{a,a'}$	$= \begin{cases} 1 & \text{if aircraft } a \text{ is assigned to runway } r \\ & \textit{and} \text{ aircraft } a' \text{ is assigned to runway } r' \\ 0 & \text{otherwise} \end{cases}$	$\forall a \neq a' \in A,\ r, r' \in R$

$L_a + Sep^{r,r'}_{w(a),w(a')} - T_{a'}$ (Beasley et al., 2000). Eq. (6) ensures that each aircraft is assigned to exactly one runway. Eq. (7) ensures that the assigned runway is allowed to handle operations of the respective aircraft class. Eq. (8) sets the $z^{r,r'}_{a,a'}$ variable: If aircraft a is assigned to runway r and aircraft a' ($a' \neq a$) is assigned to runway r', then the respective variable, $z^{r,r'}_{a,a'}$, is set to 1.

4. Dynamic programming approach

In this section, we describe a dynamic programming approach that derives optimal solutions for the ASP, as defined in Section 3. Existing DP approaches cannot be applied, as they are restricted to a single runway or to successive separation and identical independent runways (see Section 2.2.3). To consider complete separation and interdependent runways, we modify the approach by Lieder et al. (2015) as follows:

- The *state definition* is expanded and holds more information on scheduled operations that is necessary to ensure complete and diagonal separation.
- The *state transition* function has to be adjusted to handle the additional information in the state definition.
- The performance-enhancing *dominance rule* has to consider the heterogeneity of the runways when eliminating non-optimal states.

4.1. State definition

Each state s of the DP's state space $\mathcal{S}$ represents a set of feasible (partial) runway schedules with common properties. A state s is defined by a vector $(k_1, \ldots, k_{|W|})$ and a matrix $\mathcal{P} = (p_{wr})$:

$$s = \left(k_1, \ldots, k_{|W|}, \begin{pmatrix} p_{11} & \cdots & p_{1|R|} \\ \vdots & \ddots & \vdots \\ p_{|W|1} & \cdots & p_{|W||R|} \end{pmatrix} \right) \tag{9}$$

Vector $(k_1, \ldots, k_{|W|})$ indicates how many operations of each operation class are already scheduled. If we assume 3 aircraft classes, the vector consists of 6 elements (3 for take-offs and 3 for landings). Note that, from the information on how many operations of each class have been scheduled, we can derive which operations have been scheduled because we assume FCFS within each operation class.

Matrix $\mathcal{P}$ contains the times of the latest runway operations for each operation class $w \in W$ on each runway $r \in R$. If no operations of class w are scheduled on runway r, we indicate this by inserting a dummy operation with no separation requirements and with start time $p_{wr} = -1$. Accordingly, the initial state of the DP is $s_0 = (0^{|W|}, -1^{|R| \cdot |W|})$, *i.e.*, no operations have been scheduled and all runways are empty.

For each state $s \in \mathcal{S}$, we can use matrix $\mathcal{P}$ and the separation times, $Sep^{r,r'}_{w,w'}$, to derive a runway availability matrix, $\mathcal{AV}$, with the earliest possible time for the next operation of each aircraft class on each runway. To ensure complete and diagonal separation when calculating the matrix entries of $\mathcal{AV}$, we have to take the last scheduled operations of all classes on all runways into account, except for dummy operations with $p_{wr} = -1$:

$$av_{wr} = \max_{w',r'} \{p_{w'r'} + Sep^{r',r}_{w',w} \mid p_{w'r'} \geqslant 0\} \quad \forall w \in W,\ \forall r \in R \tag{10}$$

Note that av_{wr} is always set to values of at least $\max_{w',r'}\{p_{w',r'}\}$, even if we assume independent runways. In this manner, we enforce a forward-directed scheduling since no operation can be scheduled earlier than the latest operation scheduled so far. If operations of class w cannot be performed on runway r, we set the respective runway availability time, av_{wr}, to $+\infty$. Each state s is associated with a cost, $Z(s)$, that is the minimum total delay penalty cost of all feasible (partial) runway schedules represented by s.

4.2. State transition

A state transition $s \xrightarrow{(a,r)} s'$ corresponds to scheduling an aircraft a with operation class $w(a)$ on runway r. As the optimization goal is to minimize the total delay penalty cost, we schedule a as early as possible, that is, either at the beginning of its respective time window T_a or as soon as runway r becomes available. C_a, the operation start time of aircraft a, can be derived as follows:

$$C_a = \max\{T_a;\ av_{w(a),r}\}, \tag{11}$$

State $s' = (k'_1, \ldots, k'_{|W|}, \mathcal{P}')$ is then defined as:

- Number of scheduled operations per class:
  $k'_{w(a)} = k_{w(a)} + 1$,
  $k'_{w'} = k_{w'}$ for all $w' \in W \setminus w(a)$.
- Time of the last operation of all classes on all runways:
  $p'_{w(a),r} = C_a$,
  $p'_{w'r'} = p_{w'r'}$ for all $p'_{w'r'} \in \mathcal{P} \setminus p'_{w(a),r}$.

Given this information, we can then update the runway availability matrix, $\mathcal{AV}$, of s' using Eq. (10). Because we assume FCFS within each aircraft class, the aircraft a to be scheduled is always the $k_w + 1^{st}$ aircraft of class $w(a)$. Each state has at most $|W| \cdot |R|$ outgoing transitions. The state transition is associated with costs $c(w(a), C_a - T_a)$ for the additional scheduling of aircraft a on runway r at time C_a. There is no transition if the time window of a is violated, *i.e.*, $C_a > L_a$. In this manner, we also ensure that runway r can handle the respective operation class, because otherwise $av_{w(a),r} = \infty \ \Rightarrow\ C_a > L_a$.

We define set $\mathcal{S}^{(s',a,t)} \subset \mathcal{S}$ as the set of states for which a transition exists into state s' by scheduling operation a at time t. According to Bellman's Principle of Optimality (Bellman, 1957), the cost $Z(s')$ of each state $s' \in \mathcal{S}$ can be calculated recursively as the minimum cost of any state $s \in \mathcal{S}^{(s',a,t)}$ and the additional cost of the respective transition $s \xrightarrow{(a,r)} s'$:

$$Z(s') = \begin{cases} \min\limits_{\mathcal{S}^{(s',a,t)}} \{Z(s) + c(w(a), C_a - T_a)\} & \text{if } \sum\limits_{w \in W} k_w(s') \geqslant 1 \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

The only state s with $\sum_{w \in W} k_w(s) = 0$, *i.e.*, with no scheduled runway operations, is the initial state, s_0, with no delay costs and with no predecessors. The subset of states $\mathcal{S}^{final} \subset \mathcal{S}$ is defined as all states with $\sum_{w \in W} k_w(s) = |A|$, *i.e.*, with all operations scheduled. These states represent *feasible solutions* to the respective ASP instance. An *optimal solution* to the ASP is then the minimum-cost schedule represented by any state s^* with

$$s^* = \arg\min_{s \in \mathcal{S}^{final}} \{Z(s)\} \tag{13}$$

We can derive the respective minimum-cost schedule represented by s^* by tracking all transitions from the initial state s_0 to s^*.

A straightforward implementation of the recursion would result in excessive computation times due to the large set of possible predecessors $\mathcal{S}^{(s',a,t)}$ of each state s'. As, on the other hand, each state has at most $|W| \cdot |R|$ outgoing transitions, we purposefully build and traverse the state space iteratively instead of readily applying Eq. (12). We organize all states in levels, $l = 0, \ldots, |A|$, ordered by the total number of scheduled runway operations. The only state on level $l = 0$ is the initial state, s_0. States on level $l = |A|$ represent feasible schedules and the states on this level with minimal $Z(s)$ are s^*.

4.3. Dominance rule

For the ASP with successive separation and independent identical runways, Lieder et al. (2015) describe a dominance rule that significantly reduces their state space while maintaining optimality. The basic idea of this rule is, that a state of the DP can be removed, if there exists another state with:

- At least the same number of operations of all classes that have been scheduled,
- all runways are available for all operation classes at the same time or earlier, and
- the total penalty cost is the same or lower.

To apply this rule to the generalized DP presented in this paper, we have to make changes to consider runway settings with heterogeneous and interdependent runways:

If we have multiple runways that allow the same classes of runway operations and share the same interdependence with other runways, then there exist symmetric schedules, *i.e.,* schedules that are identical except that the runway assignments of two or more (identical) runways are different. To detect and prune these symmetric schedules, we divide the set of runways, R, into a set of symmetric runways, R^{sym}, and into a set of asymmetric runways, R^{asym}. In any feasible schedule, the order of symmetric runways can be changed arbitrarily.

When searching for dominated states, we always consider pairs of states, s and s'. We say that *s dominates s'* if the following criteria are met:

1. State s has at least the same number of operations scheduled for all classes as state s':
   $k_w(s) \geqslant k_w(s')$ for all $w \in W$.

2(a). In state s, all asymmetric runways are available for all classes of runway operations not later than in state s':
   $av_{w,r}(s) \leqslant av_{w,r}(s')$ for all $w \in W,\ r \in R^{asym}$.

2(b). Over all symmetric runways $(r_1, \ldots, r_k) \in R^{sym}$, there exists a permutation $\sigma(R^{sym})$ so that each symmetric runway, r_i, in state s is available for each class of operations not later than the respective runway $r'_{\sigma(i)}$ in state s':
   $\exists\ \sigma(R^{sym}) : av_{w,r}(s) \leqslant av_{w,\sigma(r)}(s')$ for all $w \in W,\ r \in R^{sym}$.

3. The cost associated with s is not higher than the cost associated with s':
   $Z(s) \leqslant Z(s')$.

Theorem 1. *If state s' is dominated, it can be removed from further consideration while still ensuring optimality. No final state that s' could be transformed into can have a lower total cost than the best final state that s could be transformed into.*

Proof. Let $Sc(s)$ and $Sc(s')$ be partial runway schedules represented by states s and s', respectively. Let Sc' be any partial runway schedule that creates a feasible solution to the respective problem instance if appended to $Sc(s')$, *i.e.,* in $(Sc(s') \circ Sc')$, all operations are scheduled and all separation constraints are met. Due to criterion 2, Sc' can always be appended to $Sc(s)$ as well, without violating any separation constraints. Due to criterion 1, in $(Sc(s) \circ Sc')$, all runway operations are scheduled as well. If $k_w(s) > k_w(s')$ for any w, one or more operations are scheduled twice in $(Sc(s) \circ Sc')$, but duplicate operations can be removed without increasing the schedule's cost. Due to criterion 3, the cost of $(Sc(s) \circ Sc')$ is not higher than the cost of $(Sc(s') \circ Sc')$, *i.e.,* if $(Sc(s') \circ Sc')$ is an optimal solution, so is $(Sc(s) \circ Sc')$. Therefore, s' can always be removed in favor of s while maintaining optimality. □

Concerning criterion (2.b): as the order of the symmetric runways can be changed arbitrarily, it is sufficient to find one permutation of these runways for which the criterion holds, analogous to criterion (2.a) for asymmetric, *i.e.,* non-interchangeable, runways. The number of permutations to consider is factorial in $|R^{sym}|$, but as this is usually a small number, the achieved state space reduction outweighs this effort.

Table 5 shows a pair of states, s and s', to exemplify the application of the dominance rule. This example assumes two symmetric, interdependent runways and separation times as in Table 1. The runway availability $\mathcal{AV}(s)$ is better than $\mathcal{AV}(s')$, because $av_{w,r}(s) \leqslant av_{w,\sigma(r)}(s')$ holds for the permutation $\sigma = (r_2, r_1)$, *i.e.,* criterion 2.b) holds in favor of s. Criteria 1 and 3 hold as well because $k(s) = k(s')$ and $Z(s) < Z(s')$, *i.e.,* both states have the same operations scheduled and the cost of s is lower than the cost of s'. Therefore, s dominates s' and s' can be removed from the state space.

Without this dominance rule, the state space of the DP becomes prohibitively large, even for small problem instances. However, to detect and remove dominated states, a large number of comparisons between pairs of states has to be performed. Therefore, it is necessary to choose a data structure that makes it possible to efficiently retrieve all states that are candidates for being dominated.

Table 5
Example application of the dominance rule.

State s:

$$Z = 1686$$
$$k = (1,1,0,0,4,0)$$
$$\mathcal{P}^T = \begin{pmatrix} -1 & 623 & -1 & -1 & 563 & -1 \\ 663 & -1 & -1 & -1 & 573 & -1 \end{pmatrix}$$
$$\mathcal{AV}^T = \begin{pmatrix} 754 & 703 & 703 & 698 & 698 & 698 \\ 745 & 732 & 723 & 738 & 738 & 738 \end{pmatrix}$$

Partial schedule:

r	a	$w(a)$	T_a	C_a
1	1	TL	0	0
1	10	TL	363	363
1	15	TL	563	563
1	2	LL	8	623
2	16	TL	573	573
2	12	LS	435	663

State s':

$$Z = 1868$$
$$k = (1,1,0,0,4,0)$$
$$\mathcal{P}^T = \begin{pmatrix} 754 & 623 & -1 & -1 & 563 & -1 \\ -1 & -1 & -1 & -1 & 573 & -1 \end{pmatrix}$$
$$\mathcal{AV}^T = \begin{pmatrix} 836 & 823 & 814 & 829 & 829 & 829 \\ 794 & 794 & 794 & 754 & 754 & 754 \end{pmatrix}$$

Partial schedule:

r	a	$w(a)$	T_a	C_a
1	1	TL	0	0
1	10	TL	363	363
1	15	TL	563	563
1	2	LL	8	623
1	12	LS	435	754
2	16	TL	573	573

Operation classes: LS (small landing), LL (large landing), TL (large take-off)

5. Rolling planning horizon heuristic

For large problem instances, the state space can become very large and, consequently, difficult to solve to optimality within short computation times. A promising heuristic solution approach is to decompose the problem into a set of smaller problems with shorter planning horizons and less runway operations. These smaller problems are then iteratively solved. The approach has two main parameters, the scheduling time window size (TW) and step size ($step$), and two auxiliary parameters, the start and end of the time window (TW_{start} and TW_{end}). It can be described as follows.

1. We initialize the time window under consideration:
   $TW_{start} \leftarrow 0$ and $TW_{end} \leftarrow TW$. There are no fixed runway operations.
2. We use DP to solve the problem. Only aircraft with target times within the current time window ($TW_{start} \leqslant T_a \leqslant TW_{end}$) are scheduled. If we have fixed operations from previous iterations, these operations define the initial state, and therefore the runway availability, of the current iteration. This way, we ensure that all separation constraints between two consecutive iterations are upheld. From the resulting runway schedule, we fix all operation with $C_a \leqslant (TW_{start} + step)$.

3(a). If we have aircraft that were not considered in the current or a previous iteration ($TW_{end} < \max_{a \in A} T_a$), we shift the time window ($TW_{start} \leftarrow (TW_{start} + step)$ and $TW_{end} \leftarrow (TW_{start} + TW)$) and go back to step 2.
3(b). Else, we have a runway schedule for all aircraft and stop iterating.

The RPH heuristic can neither guarantee to find an optimal nor a feasible solution. For the performance of RPH approaches, the right choice of the parameters TW and $step$ is essential (Stolletz and Zamorano, 2014). We examine different combinations of TW and $step$ values in our numerical study (Section 6.4). Xiangwei et al. (2011) propose a similar approach for interdependent runways, scheduling a fixed number of aircraft in each iteration using a standard MIP solver to iteratively solve the subproblems. Furini et al. (2015) evaluate different strategies for finding favorable subproblems in a single-runways setting, using a standard MIP solver as well as a tabu search heuristic to solve the subproblems. Rolling horizon approaches for the ASP with approach paths are given by Samà et al. (2013, 2014a,b).

6. Numerical study

6.1. Generation of problem instances

For the numerical evaluation of the proposed solution approach, we generated a large set of problem instances under realistic assumptions. The problem instances have exactly 50% take-offs and 50% landings, *i.e.,* we always have an even

Table 6
Distribution of operation classes ($w(a)$).

Aircraft class		Operation type			
		Landing	50%	Take-off	50%
Small	20%	Small landing	10%	Small take-off	10%
Large	40%	Large landing	20%	Large take-off	20%
Heavy	40%	Heavy landing	20%	Heavy take-off	20%

number of operations. We assume three aircraft weight classes and a distribution of the resulting $|W| = 6$ operation classes as in Table 6. This distribution is realistic for traffic at major airports (Balakrishnan and Chandran, 2010).

The numbers of runway operations, $|A|$, in each problem instance depends on the runway load, RL (in number of operations/hour), and on the length of the planning horizon, PH (in seconds). We generate samples for demand of landings and take-offs according to a Poisson process (see, *e.g.,* Willemain et al., 2004; Daniel and Harback, 2009). For each sample of inter-arrival times, we generate 10 problem instances. We build a set of target times, T_a, by randomizing the given inter-arrival times, and we randomly assign operation classes, $w(a)$, to these target times. For each set of instances, the number of operations per class and the target time of the last operation are the same.

As landing delays are more expensive than take-off delays, we assume delay costs of one monetary unit per second for take-offs and two monetary units per second for landings. We set the latest feasible operation time, L_a, to $T_a + 900$ s for all operations. For pairs of operations on the same runway, we assume separation times as in Table 1a. For interdependent runways, we assume a class-independent separation time of 40 s between two landings and no separation requirements between two take-offs or between a take-off and a landing (see Table 1b). These separation times were derived from the official separation distances issued by the FAA (Balakrishnan and Chandran, 2010; De Neufville and Odoni, 2013).

In the following, we first analyze the computational performance of the DP approach using a realistic basic problem setting with two interdependent runways. Many major airports operate two interdependent runways, *e.g.,* Mexico City International Airport (MEX) or Dubai International Airport (DXB). We derive optimal runway schedules for different numbers of runway operations and planning horizons (Section 6.2). Then, we apply the DP approach to the runway configurations of London-Heathrow (LHR) and Frankfurt Airport (FRA). We analyze, how optimized runway schedules can be used to gain additional runway capacity compared to an FCFS scheduling discipline (Section 6.3). In Section 6.4, we analyze the performance of the RPH heuristic with respect to total delay cost and computation time, comparing different parameter settings. Detailed results for all problem instances are given in the Appendix A.

All calculations were performed on a Windows 7 desktop PC with an Intel i7-5820k CPU and 32 GB DDR4-RAM. The DP approaches were implemented in Java 1.8.0; the MIP was implemented in GAMS 24.1 and solved with CPLEX 24.4.1. The problem instances can be downloaded from http://stolletz.bwl.uni-mannheim.de/en/library.

6.2. Performance analysis of the exact DP approach

In this section, we analyze the computational performance of the DP approach. As a benchmark, we use the MIP defined in Section 3 with a time limit of 60 min. As a basic problem setting, we consider a two-runway system with interdependent runways, a runway load of $RL = 90$ operations/hour, and a planning horizon of $PH = 1800$ s. Ninety operations/hour is a realistic load for peak hours on a fully utilized two-runway system, *e.g.,* Munich Airport (MUC) in Germany (89 operations/hour, Koesters, 2007) and London-Heathrow (88 operations/hour, ACL, 2015).

Table 7 shows the aggregated computational results for the basic problem case. To assess the impact of the runway load on computation time and delays, we also solve instances with a reduced load of $RL = 80$ and with an increased load of $RL = 100$. To assess the impact of runway interdependence, we also solve the same instances on two independent runways. For each setting, we solve 10 problem instances with both MIP and DP and we report average and maximum computation times, average and maximum delays for all take-offs, landings, and the last scheduled operations. We also report two metrics based on *position shifts*. A position shift is defined as the absolute difference between the position of an operation in the optimized sequence and its position in the respective FCFS sequence. We report the maximum position shifts to both earlier and later positions and the average sum of position shifts (which is, by definition, the same for both directions). For the MIP, we also report the gaps to the (optimal) DP solution and to the lower bound after the 1 h time limit was hit. Detailed results of the individual instances aggregated in Table 7 are given in the Appendix in Tables 13 and 14.

For $RL = 80$ and $RL = 90$, the DP returned optimal results in less than one minute for all instances; for $RL = 100$, it returned optimal results in less than 4 min on interdependent runways and in less than 18 min on independent runways. The computation times of both MIP and DP increase with increasing runway load. The MIP was able to find proven optimal results only for some instances with a low load of $RL = 80$. For instances with higher load, it always

Table 7
Basic problem cases: impact of runway load.

Runway configuration			Interdependent			Independent		
Runway load (operations/hour)			$RL = 80$	$RL = 90$	$RL = 100$	$RL = 80$	$RL = 90$	$RL = 100$
Runway operations			40	46	50	40	46	50
DP	CPU times	Average (seconds)	7	16	56	7	17	131
		Maximum (seconds)	9	29	217	11	40	934
	Delays (seconds)	Take-off (avg./max.)	27/302	46/398	68/605	31/274	43/331	74/424
		Landing (avg./max.)	27/258	39/484	58/384	17/246	29/401	37/227
		Last operation (avg./max.)	76/302	63/275	142/316	70/274	62/291	121/300
	Position shifts	Avg. sum of pos. shifts	9.7	18.1	23.8	10.8	17.0	28.7
		Max. shift (earlier/later pos.)	3/6	4/11	7/15	3/9	4/9	8/9
MIP w. CPLEX	CPU times	Average (seconds)	2757	3600	3600	2739	3600	3600
		Maximum (seconds)	3600	3600	3600	3600	3600	3600
	Opti- mality	Gap/Opt. (avg./max.; percent)	0.7/5.4	5.9/21.3	14.0/21.6	0.4/2.0	3.5/12.8	11.6/26.6
		Gap/LB (avg./max.; percent)	12.2/49.2	56.7/79.3	83.6/94.9	13.7/43.5	47.5/79.7	79.3/95.6
		Optimum found	7/10	3/10	0/10	8/10	3/10	0/10
		Optimum proven	4/10	0/10	0/10	4/10	0/10	0/10
	Delays (seconds)	Take-off (avg./max.)	28/251	47/309	77/477	31/288	43/296	75/396
		Landing (avg./max.)	27/258	43/519	66/377	17/246	31/298	47/373
		Last operation (avg./max.)	59/143	64/275	139/261	71/288	57/275	129/330
	Position shifts	Avg. sum of pos. shifts	9.3	18.0	27.5	10.9	15.2	25.3
		Max. shift (earlier/later pos.)	3/6	5/10	6/11	3/6	5/5	5/8

hit the time limit of one hour. In most of these instances, the best solution found by the MIP is not optimal. The gaps for both the optimal objective value and lower bound increase with increasing runway load. High runway loads lead to higher average delays for both take-offs and landings. As take-offs are cheaper to delay than landings, the average take-off delay is higher than the average landing delay. Compared to the interdependent runways setting, the independent runway setting shows higher average computation times and lower average delays. The number of position shifts and the maximum shifts to earlier or later positions increase with increasing runway load. As position shifts are neither limited nor penalized, the maximum position shift takes high values of up to 15 positions. However, no operation is delayed more than 605 s.

Fig. 1 shows the first of ten problem instances of the basic problem setting with $RL = 90$ operations/hour. The top row shows the target times and operation classes of all runway operations. Below, we have optimal runway schedules for independent as well as interdependent runways. Differences between both schedules are highlighted. Due to the diagonal separation constraints on interdependent runways, some operations have more delay than on independent runways. These additional delays have an impact on all successive operations unless there is runway idle time to "absorb" these delays. Thus, the optimal schedules considerably differ towards the end of the planning horizon.

To assess the impact of the planning horizon and, therefore, the number of runway operations on computation time and delays, we solve our basic problem ($RL = 90$ operations/hour, 2 interdependent runways) with different planning horizons of $PH = 1200, 1500, 1800, 2100$, and 2400 s. The results are shown in Table 8. Detailed results are given in

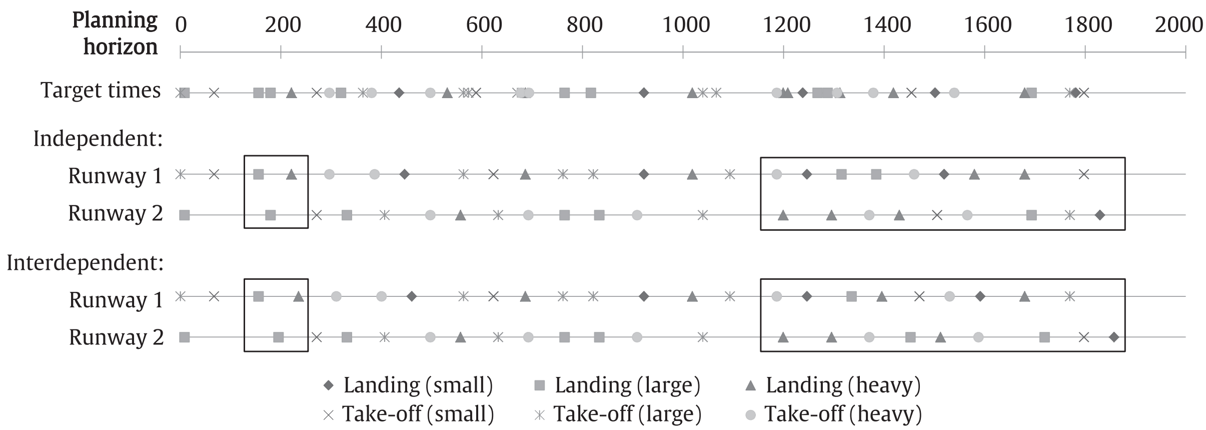

Fig. 1. Example of a basic problem instance with solutions.

Table 8
Basic problem cases: Impact of planning horizon.

Planning horizon (seconds)			$PH = 1200$	$PH = 1500$	$PH = 1800$	$PH = 2100$	$PH = 2400$
Runway operations			30	40	46	54	60
DP	CPU times	CPU time (avg.; seconds)	2	8	16	42	58
		CPU time (max.; seconds)	5	14	29	94	80
	Delays (seconds)	Take-off (avg./max.)	32/287	37/268	46/398	45/477	54/399
		Landing (avg./max.)	36/312	32/364	39/484	50/364	45/256
		Last operation (avg./max.)	60/192	75/224	63/275	68/193	60/158
	Position shifts	Avg. sum of pos. shifts	9.5	13.3	18.1	19.1	26.2
		Max. shift (earlier/later pos.)	3/7	3/8	4/11	7/12	5/10
MIP w. CPLEX	CPU times	CPU time (avg.; seconds)	3051	3600	3600	3600	3600
		CPU time (max.; seconds)	3600	3600	3600	3600	3600
	Opti- mality	Gap/Opt. (avg./max.; percent)	0.1/1.0	3.1/7.1	5.9/21.3	8.7/17.0	8.6/17.1
		Gap/LB (avg./max.; percent)	14.6/39.7	44.8/61.7	56.7/79.3	72.7/87.5	75.3/89.3
		Optimum found	9/10	3/10	3/10	0/10	0/10
		Optimum proven	2/10	0/10	0/10	0/10	0/10
	Delays (seconds)	Take-off (avg./max.)	32/237	38/256	47/309	46/272	60/385
		Landing (avg./max.)	37/276	34/236	43/519	56/385	49/286
		Last operation (avg./max.)	60/192	74/214	64/275	80/248	94/238
	Position shifts	Avg. sum of pos. shifts	9.5	12.4	18.0	22.1	25.1
		Max. shift (earlier/later pos.)	3/6	3/6	5/10	5/6	7/9

Table 15 in the Appendix. The computation times of the DP increase for larger planning horizons. However, even for a planning horizon of $PH = 2400$ and 60 runway operations, the DP returns optimal results in less than 2 min, while the MIP hits the time limit in all instances with $PH = 1500$ or higher. The MIP returns proven optimal solutions for only 2 out of 10 instances with $PH = 1200$. For longer planning horizons, only for some instances were unproven optimal solutions found. The optimality gaps of the MIP, the average delays, and the position shift metrics increase with increasing planning horizon.

To summarize, the detailed results in the Appendix (Tables 13–15), that refer to Tables 7 and 8 of this section, show that the DP significantly outperforms the MIP in all instances.

6.3. General runway configurations

In this section, we analyze two runway settings from two of Europe's busiest airports, London-Heathrow (LHR) and Frankfurt Airport (FRA).

We use the DP approach to calculate optimal runway schedules with a planning horizon of $PH = 1800$. As a benchmark for the quality of the optimized runway schedules, we use an FCFS scheduling rule, as it is applied in practice (Bianco et al., 2006; De Neufville and Odoni, 2013). We optimize the assignment of operations to runways, and only the sequence of operations is given. To derive an FCFS schedule with the DP approach, we allow state transitions ($s \xrightarrow{(a,r)} s'$) only if aircraft a has the smallest target time, T_a, of all unscheduled aircraft. To derive an FCFS schedule with the MIP, we fix the sequence of operations by adding the constraint: $\delta_{a,a'} = 1 \; \forall a, a' \in A, T_a < T'_a$. We also analyze the additional runway capacity gained by optimized runway schedules by using the same set of problem instances and inserting additional runway operations (a large take-off with a target of $T_a = 600$ s and a large landing with a target of $T_a = 1200$ s).

LHR has two independent runways, one of which is operated in mixed mode while the other is used only for landings. This practice reduces the noise exposure of nearby residents, but it incurs a significant reduction of capacity (Irvine et al., 2015). We assume a runway load of $RL = 90$ operations/hour, *i.e.*, we have 46 operations in the planning horizon.

Table 9 shows objective values, computation times, and delays for the LHR setting. All instances were solved in 24 s or less. Compared to FCFS schedules, the average total weighted delay penalty can be reduced to less than one third. All delay figures of the optimized schedules are lower than those of the FCFS schedules, except for the maximum take-off delay. Even if we compare the optimized schedules with two additional operations to the FCFS schedules without additional operations, the objective value and most delay figures are significantly smaller.

FRA has four runways. Runway 1 is used only for landings, and it is too short to be used by heavy aircraft, such as the Airbus A380. Runway 2 is only used for take-offs due to noise considerations and safety issues. Runways 3 and 4 are used for take-offs and landings, but as they are parallel and close to each other (550 m), they have to be treated as a single runway, except for the sequence "landing, then take-off," which can be performed without separation time (see Table 1b). The

Table 9
LHR's runway setting ($RL = 90$ operations/hour).

Solution approach		DP (exact)		DP (FCFS)	
Runway operations		46	46 + 2	46	46 + 2
Solution	Objective value (average)	5403	6548	16630	20477
	Objective value (maximum)	7347	8478	22628	28325
	CPU time (average, seconds)	13	19	<1	<1
	CPU time (maximum, seconds)	17	24	<1	<1
Delays (seconds)	Take-off (average)	138	169	258	298
	Take-off (maximum)	797	825	722	787
	Landing (average)	47	51	232	277
	Landing (maximum)	508	508	690	832
	Last operation (average)	326	374	489	569
	Last operation (maximum)	463	488	676	736

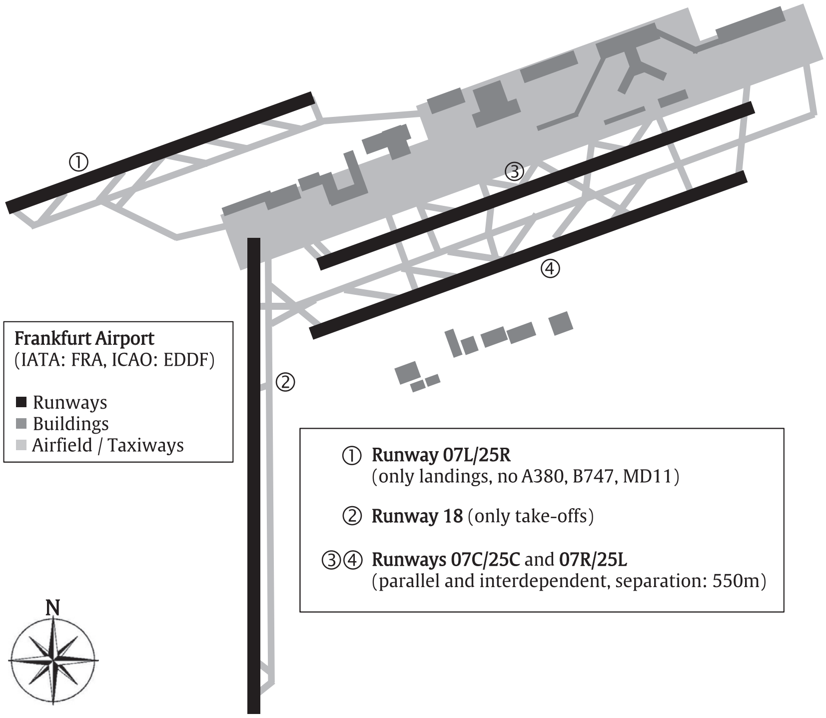

Fig. 2. Schematic representation of Frankfurt Airport (Germany).

Table 10
FRA's runway setting ($RL = 120$ operations/hour).

Solution approach		DP (exact)		DP (FCFS)	
Runway operations		60	60 + 2	60	60 + 2
Solution	Objective value (average)	3563	3966	12932	14396
	Objective value (maximum)	6683	7492	23468	26587
	CPU time (average, seconds)	542	745	<1	<1
	CPU time (maximum, seconds)	1040	1391	<1	<1
Delays (seconds)	Take-off (average)	21	22	128	136
	Take-off (maximum)	179	179	452	495
	Landing (average)	48	52	151	163
	Landing (maximum)	341	490	498	538
	Last operation (average)	111	122	264	283
	Last operation (maximum)	242	242	452	475

Table 11
Rolling planning horizon heuristic for FRA's runway setting.

RPH	*TW* parameter (seconds)	600	900	900	1200	1200	1200	
	step parameter (seconds)	300	300	600	300	600	900	Exact
	Time window overlap (seconds)	300	600	300	900	600	300	solution
	Iterations	5	4	3	3	2	2	
60 operations	Objective value (average)	3570	3563	3563	3563	3563	3570	3563
	Objective value (maximum)	6683	6683	6683	6683	6683	6683	6683
	Gap/Opt. (average; percent)	0.2	0.0	0.0	0.0	0.0	0.2	0.0
	Gap/Opt. (maximum; percent)	1.6	0.0	0.0	0.0	0.0	1.6	0.0
	Optimum found	9/10	10/10	10/10	10/10	10/10	9/10	10/10
	CPU time (average, seconds)	36	190	65	514	220	194	542
	CPU time (maximum, seconds)	66	359	130	1277	459	436	1040
60 + 2 operations	Objective value (average)	3979	3966	3969	3966	3966	3976	3966
	Objective value (maximum)	7494	7492	7494	7492	7492	7492	7492
	Gap/Opt. (average; percent)	0.4	0.0	0.1	0.0	0.0	0.3	0.0
	Gap/Opt. (maximum; percent)	1.6	0.0	0.6	0.0	0.0	1.6	0.0
	Optimum found	6/10	10/10	8/10	10/10	10/10	8/10	10/10
	CPU time (average, seconds)	49	270	89	672	313	217	745
	CPU time (maximum, seconds)	86	578	229	1703	764	463	1391

runway layout of FRA is shown in Fig. 2. We assume a runway load of $RL = 120$ operations/hour, *i.e.,* we have 60 runway operations.

Table 10 shows objective values, computation times, and delays for the FRA setting. The computation time increases up to 24 min in one instance. Similar to the results of the LHR setting, the average total weighted delay penalty of the optimized schedules is reduced to less than a third compared to the respective FCFS schedules. All delay figures of the optimized schedules are lower than those of the FCFS schedules, even if we compare optimized schedules with two additional operations to FCFS schedules without additional operations.

In conclusion, for both runway settings, optimized runway schedules make it possible to schedule additional runway operations while reducing delays significantly compared to FCFS runway schedules. However, if computation times are too long for practical use, a heuristic approach should be used.

6.4. Rolling planning horizon heuristic

For some instances of the FRA problem setting, the exact DP approach takes several minutes of computation time, which is not fast enough for practical use. In Section 5, we proposed an RPH heuristic. Table 11 shows objective values, optimality gaps, and computation times of this heuristic for the FRA problem instances. We analyze different time window sizes, *TW*, of 600, 900, and 1200 s and different *step* sizes of 300, 600, and 900 s.

All analyzed combinations of *TW* and *step* yield very good results in terms of optimality. 110 out of 120 problem instances are solved to optimality. For an overlap, *i.e.,* the difference between *TW* and *step*, of 600 s or more, the heuristic returns optimal solutions for all instances. For a smaller overlap of 300 s, the gap to the optimal solution is 1.6% or less.

There is a trade-off between solution quality and computation times when setting the *TW* and *step* parameters of the heuristic. On one hand, a big *TW* and a small *step* parameter result in large subproblems and a large overlap and therefore favor high-quality results. On the other hand, large subproblems require longer computation times and a large overlap results in an increased number of iterations. In some instances with, *e.g., TW* $= 1200$ and *step* $= 300$ (column 4), this may result in longer computation times than the exact approach. The combination *TW* $= 900$ s and *step* $= 600$ s (column 3) yields close-to-optimal results in fast computation times.

7. Consideration of costs for taxiing

The ASP discussed in this paper can be extended to consider taxi costs, $c^{taxi}(a, r)$, for taxiing an aircraft, a, from its gate to its assigned runway, r, or vice versa. The objective function (Eq. 1) of the MIP has to be modified as follows.

$$\text{Minimize} \quad F = \underbrace{\sum\nolimits_{a \in A} c^{delay}(w(a), C_a - T_a)}_{\text{delay penalties}} + \underbrace{\sum\nolimits_{a \in A} \sum\nolimits_{r \in R} c^{taxi}(a, r) \cdot y_a^r}_{\text{taxi costs}} \tag{14}$$

Table 12
Impact of taxi costs.

Runway setting		DP		MIP with CPLEX			
		Obj. value	CPU (s)	Obj. value	CPU (s)	Gap (DP)	Gap (LB)
Inter-dependent	Average	5030	31	5298	3600	4.6%	30.1%
	Maximum	8126	55	8418	3600	13.9%	50.0%
Independent	Average	4475	35	4599	3600	2.2%	19.9%
	Maximum	6867	75	7397	3600	7.2%	44.0%

Considering taxi costs result in optimal schedule which does not satisfy the FCFS condition within each operation class. Therefore, constraint (4) of the MIP has to be removed. To consider taxi costs in the DP approach, the Bellman recursion (Eq. 12) has to be modified as follows.

$$Z(s') = \begin{cases} \min\limits_{S^{(s',a,t)}} \quad \{Z(s) + c^{delay}(w(a), C_a - T_a) + c^{taxi}(a,r)\} & \text{if } \sum\limits_{w \in W} k_w(s') \geqslant 1 \\ 0 \quad \text{otherwise} \end{cases} \tag{15}$$

In addition, when applying the dominance rule, runways with different taxi costs are defined as *asymmetric*. As the DP approach always derives schedules with FCFS within each class, we cannot guarantee optimal results when considering taxi costs.

To assess the impact of taxi costs, we generate a set, $c^{taxi}(a,r)$, of taxi costs using a uniform distribution in the interval $[0, 100]$. We solve the 10 instances of the basic problem setting on both interdependent and independent runways under consideration of taxi costs. Detailed results are given in the Appendix (Table 16) and are summarized in Table 12.

Although we cannot guarantee optimal results for the DP approach, the MIP did not find a solution with better objective value for any instance within the one-hour time limit. As the two runways are no longer symmetric, the computation times of the DP are approximately twice as high as for the basic case without taxi costs. The optimality gaps of the MIP are better than for the basic case because the taxi costs help increasing the respective lower bounds.

8. Conclusion and outlook

In this paper, we analyze the *aircraft scheduling problem* (ASP) for heterogeneous and interdependent runways. We define the problem as a MIP and provide an efficient DP approach to optimally solve the ASP. To cope with large problem instances, we present an RPH heuristic. We demonstrate the high computational performance of both approaches in a numerical study. The RPH heuristic returns close-to-optimal results in very short computation times. Using realistic runway settings, we show that additional runway capacity can be gained from optimized runway schedules compared to FCFS schedules.

For further research, the airport's TMA and ground operations could be considered, that is, to integrate approach paths, taxiway routing (Marín, 2006), and gate assignment. There is recent technical progress that will probably increase the capacity of existing runway systems: (1) there are efforts to adjust the separation requirements to the current wind situation (NATS, 2014); (2) for landings on closely spaced parallel runways, some papers present an *aircraft pairing* approach, *i.e.*, aircraft are scheduled to land in staggered pairs to avoid the other aircraft's wake vortex (Kupfer, 2009; Farrahi and Verma, 2010). The presented DP approach could also be generalized to implement other aircraft sequencing approaches like, *e.g.*, Constrained Position Shifting, or other heuristic approaches than RPH. It could also be adapted to scheduling problems from other areas of application.

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Appendix A. Detailed numerical results

See Tables 13–16.

Appendix B. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.tre.2016.01.015.

Table 13
Basic problem case (2 interdependent runways; diag. sep. 40 s.).

Inst.	DP							MIP with CPLEX								
	Obj.	CPU	Delays (s)			Pos. shifts		Obj.	CPU	Gap	Gap	Delays (s)			Pos. shifts	
	value	(s)	take-off	landing	last op.	sum	mps +/−	value	(s)	(opt.)	(LB)	take-off	landing	last op.	sum	mps +/−
(a) 40 aircraft (80 operations/hour)																
1	1839	6	20/103	35/111	0	6	1/2	1939	3600	5.4%	24.4%	24/153	36/108	0	8	2/2
2	2723	6	32/181	51/170	0	17	3/3	2755	3600	1.2%	49.2%	39/204	49/170	0	13	2/3
3	1143	8	33/251	11/70	60	11	3/6	1143	1094	0.0%	0.0%	33/251	11/70	60	12	3/6
4	1276	7	31/144	16/78	44	6	3/2	1276	3600	0.0%	10.3%	31/153	16/78	44	6	3/3
5	1895	7	34/125	30/143	143	11	3/2	1895	3600	0.0%	16.1%	34/125	30/143	143	11	3/2
6	1531	7	25/119	25/125	52	7	2/2	1531	2619	0.0%	0.0%	25/119	25/125	52	8	2/2
7	1175	6	15/72	21/93	83	3	1/2	1175	3600	0.0%	4.9%	15/72	21/93	83	3	1/2
8	1103	5	17/95	18/71	71	8	2/2	1103	2001	0.0%	0.0%	17/95	18/71	71	8	2/2
9	1655	9	41/302	20/113	302	13	2/6	1660	3600	0.3%	17.0%	40/193	21/133	139	11	2/3
10	2018	4	22/151	39/258	1	15	3/5	2018	253	0.0%	0.0%	22/151	39/258	1	13	3/5
Avg.	1636	7	27/154	27/123	76	9.7	2.3/3.2	1650	2757	0.7%	12.2%	28/152	27/125	59	9.3	2.3/3
Max.	2723	9	41/302	51/258	302	17	3/6	2755	3600	5.4%	49.2%	40/251	49/258	143	13	3/6
(b) 46 aircraft (90 operations/hour)																
1	2552	22	40/215	35/165	76	14	3/2	2552	3600	0.0%	28.2%	40/226	35/165	76	15	3/4
2	1768	13	33/137	21/158	39	9	1/3	1768	3600	0.0%	36.1%	33/117	21/158	39	10	1/3
3	2539	14	49/279	30/150	29	24	4/5	2685	3600	5.8%	71.4%	56/309	30/170	29	22	4/7
4	2130	11	48/263	21/135	36	15	3/4	2248	3600	5.5%	44.0%	33/146	32/225	36	14	2/4
5	2777	29	62/398	29/173	275	15	3/10	2869	3600	3.3%	43.0%	54/275	35/201	275	15	3/3
6	3239	13	42/182	48/243	11	21	3/4	3928	3600	21.3%	67.3%	54/182	56/247	23	17	3/5
7	2924	16	38/105	44/261	72	16	3/5	3081	3600	5.4%	67.1%	36/116	48/295	72	19	2/8
8	1704	13	40/277	16/79	20	11	2/5	1704	3600	0.0%	52.4%	40/148	16/79	20	13	2/3
9	6008	15	51/354	104/484	65	39	4/11	6584	3600	9.6%	78.2%	63/262	111/519	65	37	5/10
10	3239	11	57/337	41/167	6	17	3/6	3509	3600	8.3%	79.3%	58/290	46/192	6	18	3/4
Avg.	2888	16	46/255	39/202	63	18.1	2.9/5.5	3093	3600	5.9%	56.7%	47/207	43/225	64	18	2.8/5.1
Max.	6008	29	62/398	104/484	275	39	4/11	6584	3600	21.3%	79.3%	63/309	111/519	275	37	5/10
(c) 50 aircraft (100 operations/hour)																
1	2999	27	49/148	35/199	96	13	2/4	3232	3600	7.8%	75.2%	55/131	37/139	96	18	2/2
2	5285	43	77/436	66/265	36	27	4/7	5559	3600	5.2%	85.0%	81/445	70/363	45	35	6/10
3	8244	217	128/437	100/374	129	48	7/8	9492	3600	15.1%	91.5%	164/437	107/377	159	54	6/11
4	4824	33	69/251	61/384	207	25	4/7	5767	3600	19.5%	76.1%	65/261	82/321	261	28	3/4
5	2861	31	69/185	22/79	74	17	4/5	3156	3600	10.3%	80.4%	68/270	29/107	102	23	4/6
6	4231	42	64/299	52/180	100	21	3/6	5008	3600	18.4%	88.0%	92/390	53/319	99	32	4/6
7	4199	42	34/220	66/196	185	16	2/4	5006	3600	19.2%	82.7%	50/477	74/213	213	29	4/10
8	4274	60	97/605	36/169	316	25	4/15	4989	3600	16.7%	94.9%	70/169	64/186	143	9	2/2
9	4680	32	41/124	72/325	110	21	5/6	5689	3600	21.6%	82.4%	70/242	78/258	110	24	4/2
10	4511	29	49/192	65/240	162	25	4/4	4786	3600	6.1%	79.8%	59/251	66/205	162	23	3/7
Avg.	4611	56	68/290	58/241	142	23.8	3.9/6.6	5268	3600	14.0%	83.6%	77/307	66/249	139	27.5	3.8/6
Max.	8244	217	128/605	100/384	316	48	7/15	9492	3600	21.6%	94.9%	164/477	107/377	261	54	6/11

Table 14
Basic problem case (2 independent runways).

	DP							MIP with CPLEX								
Inst.	Obj.	CPU	Delays (s)			Pos. shifts		Obj.	CPU	Gap	Gap	Delays (s)			Pos. shifts	
	value	(s)	take-off	landing	last op.	sum	mps +/−	value	(s)	(opt.)	(LB)	take-off	landing	last op.	sum	mps +/−
(a) 40 aircraft (80 operations/hour)																
1	1413	5	38/157	16/88	0	12	2/3	1438	3600	1.8%	31.2%	39/157	16/88	0	12	2/4
2	2154	6	41/181	33/108	0	13	3/3	2198	3600	2.0%	43.5%	42/217	33/108	0	15	3/4
3	1057	8	33/251	9/68	60	11	3/6	1057	1659	0.0%	0.0%	33/251	9/68	60	12	3/6
4	1033	7	35/159	8/50	44	8	3/3	1033	3600	0.0%	10.7%	35/153	8/50	44	8	3/3
5	1687	7	34/125	24/136	136	11	3/2	1687	3600	0.0%	16.4%	34/125	24/136	136	11	3/2
6	1201	7	28/158	15/146	52	11	3/3	1201	3556	0.0%	0.0%	28/239	15/146	52	12	3/6
7	925	7	15/72	15/80	72	4	1/2	925	3600	0.0%	13.0%	15/72	15/80	72	4	1/2
8	666	5	16/73	8/58	64	5	1/1	666	229	0.0%	0.0%	16/73	8/58	59	5	1/1
9	1444	11	50/274	10/55	274	18	3/9	1444	3600	0.0%	22.2%	50/288	10/55	288	15	3/6
10	1871	5	20/85	36/246	1	15	3/5	1871	346	0.0%	0.0%	20/191	36/246	1	15	3/5
Avg.	1345	7	31/154	17/104	70	10.8	2.5/3.7	1352	2739	0.4%	13.7%	31/177	17/104	71	10.9	2.5/3.9
Max.	2154	11	50/274	36/246	274	18	3/9	2198	3600	2.0%	43.5%	50/288	36/246	288	15	3/6
(b) 46 aircraft (90 operations/hour)																
1	2154	20	37/215	28/161	48	14	3/3	2169	3600	0.7%	29.2%	39/213	27/97	48	12	2/3
2	1309	13	34/134	11/120	28	11	3/3	1309	3600	0.0%	28.4%	34/102	11/120	28	10	3/3
3	2206	15	41/285	27/170	19	19	3/5	2424	3600	9.9%	79.7%	52/234	26/159	19	19	3/5
4	1774	12	45/259	15/106	36	12	2/4	1774	3600	0.0%	38.8%	45/259	15/106	36	13	2/5
5	2449	40	57/291	24/111	291	15	3/3	2477	3600	1.1%	26.9%	57/275	24/125	275	16	3/3
6	2339	13	31/113	34/226	11	22	3/3	2352	3600	0.6%	33.5%	31/125	34/226	11	21	3/3
7	2171	16	31/145	31/141	65	10	2/3	2289	3600	5.4%	49.2%	33/83	33/234	65	10	2/4
8	1445	13	35/238	13/70	20	11	2/5	1445	3600	0.0%	55.5%	35/179	13/70	20	10	2/4
9	4751	18	65/331	70/401	65	34	4/9	5360	3600	12.8%	69.2%	46/222	93/298	65	23	5/5
10	2805	13	52/314	34/153	33	22	4/6	2944	3600	5.0%	64.8%	54/296	36/150	6	18	4/5
Avg.	2340	17	43/233	29/166	62	17	2.9/4.4	2454	3600	3.5%	47.5%	43/199	31/159	57	15.2	2.9/4
Max.	4751	40	65/331	70/401	291	34	4/9	5360	3600	12.8%	79.7%	57/296	93/298	275	23	5/5
(c) 50 aircraft (100 operations/hour)																
1	2268	27	44/107	23/110	79	14	2/3	2306	3600	1.7%	63.8%	43/107	24/110	79	12	2/3
2	4233	68	101/396	33/145	54	39	5/9	4735	3600	11.9%	84.4%	86/329	51/220	71	34	4/5
3	6936	934	191/424	42/220	132	69	8/8	8783	3600	26.6%	92.1%	143/396	104/373	185	39	5/6
4	3959	34	59/258	49/227	121	24	4/6	4519	3600	14.1%	65.8%	70/336	55/254	129	21	3/7
5	2618	32	53/197	25/83	78	17	3/7	2720	3600	3.9%	81.5%	51/206	28/92	87	14	3/4
6	3665	42	68/299	39/180	104	25	3/6	3816	3600	4.1%	83.4%	75/250	38/131	99	22	5/4
7	3118	40	24/124	50/193	193	19	2/4	3703	3600	18.8%	76.9%	51/233	48/153	153	20	2/6
8	3630	72	93/300	25/160	300	33	4/5	4272	3600	17.7%	95.6%	109/345	30/160	330	42	4/8
9	3240	27	46/383	41/185	110	24	4/9	3678	3600	13.5%	67.3%	56/262	45/202	110	24	4/5
10	3867	29	64/258	45/169	43	23	3/5	4003	3600	3.5%	82.4%	67/314	46/119	43	25	4/6
Avg.	3753	131	74/275	37/167	121	28.7	3.8/6.2	4254	3600	11.6%	79.3%	75/278	47/181	129	25.3	3.6/5.4
Max.	6936	934	191/424	50/227	300	69	8/9	8783	3600	26.6%	95.6%	143/396	104/373	330	42	5/8

Table 15
Basic problem case (2 interdependent runways; diag. sep. 40 s.).

Inst.	DP							MIP with CPLEX								
	Obj.	CPU	Delays (s)			Pos. shifts		Obj.	CPU	Gap	Gap	Delays (s)			Pos. shifts	
	value	(s)	take-off	landing	last op.	sum	mps +/−	value	(s)	(opt.)	(LB)	take-off	landing	last op.	sum	mps +/−
(a) Planning horizon: 1200 s (30 aircraft, 90 operations/hour)																
1	1809	2	36/146	42/127	16	11	2/3	1809	3600	0.0%	10.8%	36/146	42/127	16	12	2/3
2	1728	2	32/138	41/162	57	9	2/5	1746	3600	1.0%	19.8%	30/138	43/162	57	8	1/5
3	1782	2	31/208	43/151	0	9	1/4	1782	3600	0.0%	12.0%	31/208	43/151	0	9	1/4
4	1489	2	50/287	24/110	110	18	3/7	1489	3600	0.0%	39.7%	50/237	24/110	110	18	3/6
5	2441	5	56/204	53/312	28	12	2/7	2442	3600	0.0%	37.1%	57/204	53/276	28	12	2/4
6	1544	3	31/192	35/114	192	8	2/4	1544	3600	0.0%	10.0%	31/192	35/114	192	8	2/4
7	1195	2	27/171	26/81	45	7	1/2	1195	1425	0.0%	0.0%	27/171	26/81	45	7	1/2
8	1147	2	16/102	30/160	26	7	2/3	1147	3600	0.0%	7.9%	16/102	30/160	26	7	2/3
9	1747	2	27/162	44/171	62	10	2/3	1747	3600	0.0%	8.3%	27/162	44/171	62	10	2/3
10	1054	2	17/79	26/120	65	4	2/2	1054	286	0.0%	0.0%	17/79	26/120	62	4	2/2
Avg.	1594	2	32/169	36/151	60	9.5	1.9/4	1596	3051	0.1%	14.6%	32/164	37/147	60	9.5	1.8/3.6
Max.	2441	5	56/287	53/312	192	18	3/7	2442	3600	1.0%	39.7%	57/237	53/276	192	18	3/6
(b) Planning horizon: 1500 s (40 aircraft, 90 operations/hour)																
1	2294	10	50/124	32/120	91	12	3/2	2456	3600	7.1%	61.7%	55/176	34/138	109	15	3/4
2	2606	14	35/186	47/236	35	16	3/6	2728	3600	4.7%	42.6%	49/193	44/236	35	19	3/6
3	1530	9	35/150	20/90	32	6	1/2	1530	3600	0.0%	42.0%	35/150	20/90	32	6	1/2
4	1946	6	20/141	38/318	83	17	2/8	2050	3600	5.3%	48.2%	42/125	30/115	83	11	2/3
5	2001	7	43/224	28/100	224	16	2/4	2013	3600	0.6%	60.6%	44/153	28/110	153	16	3/3
6	2462	7	27/98	47/246	17	12	2/5	2542	3600	3.2%	50.0%	21/90	53/199	16	11	2/3
7	1440	4	17/134	27/122	33	9	3/2	1440	3600	0.0%	18.7%	17/134	27/122	33	9	3/2
8	1716	9	26/79	29/212	36	9	2/4	1716	3600	0.0%	10.6%	26/79	29/212	36	10	2/4
9	2371	9	64/172	27/101	172	16	3/3	2452	3600	3.4%	55.1%	51/214	36/127	214	13	2/4
10	2071	7	48/268	27/112	30	20	2/5	2217	3600	7.0%	58.2%	36/256	37/132	30	14	2/4
Avg.	2044	8	37/158	32/166	75	13.3	2.3/4.1	2114	3600	3.1%	44.8%	38/157	34/148	74	12.4	2.3/3.5
Max.	2606	14	64/268	47/318	224	20	3/8	2728	3600	7.1%	61.7%	55/256	53/236	214	19	3/6
(c) planning horizon: 2100 s (54 aircraft, 90 operations/hour)																
1	3280	35	37/122	42/203	122	8	2/3	3340	3600	1.8%	38.2%	32/122	46/203	122	20	2/3
2	3088	27	32/172	39/196	0	20	3/4	3473	3600	12.5%	71.0%	38/143	38/179	0	19	2/3
3	3861	35	39/172	53/323	34	22	2/8	4330	3600	12.1%	77.5%	50/163	56/174	34	20	2/3
4	3149	30	38/217	39/128	125	11	2/2	3426	3600	8.8%	73.4%	46/249	40/152	248	21	3/4
5	2122	26	23/74	28/119	29	11	2/2	2207	3600	4.0%	68.8%	25/89	28/128	29	12	1/2
6	2386	27	37/171	25/115	171	12	2/3	2407	3600	0.9%	66.4%	41/148	23/115	148	20	2/3
7	6158	72	93/477	67/324	193	33	4/9	7206	3600	17.0%	87.5%	79/272	93/363	221	29	5/5
8	3390	39	58/230	33/159	6	22	3/12	3680	3600	8.6%	79.5%	47/160	43/188	0	18	2/3
9	4341	35	47/450	55/210	0	21	3/7	4930	3600	13.6%	80.2%	36/156	71/288	0	31	4/6
10	7652	94	47/278	118/364	0	31	7/6	8249	3600	7.8%	84.6%	61/258	122/385	0	31	5/5
Avg.	3943	42	45/236	50/214	68	19.1	3/5.6	4325	3600	8.7%	72.7%	46/176	56/218	80	22.1	2.8/3.7
Max.	7652	94	93/477	118/364	193	33	7/12	8249	3600	17.0%	87.5%	79/272	122/385	248	31	5/6

(d) planning horizon: 2400 s (60 aircraft, 90 operations/hour)																
1	3691	48	54/246	34/167	118	31	3/7	3874	3600	5.0%	70.4%	55/238	36/187	238	24	3/6
2	3432	48	39/175	37/159	10	20	2/4	3482	3600	1.5%	65.5%	37/128	39/126	41	17	3/3
3	5280	57	44/258	65/248	44	36	3/7	5618	3600	6.4%	86.6%	61/269	63/169	44	21	3/5
4	3480	48	39/200	38/147	120	18	3/4	3773	3600	8.4%	77.2%	36/168	44/136	132	11	2/2
5	3458	63	21/108	46/225	36	16	3/4	3618	3600	4.6%	49.9%	34/236	43/273	236	16	3/5
6	3565	70	53/267	32/159	34	20	3/4	3990	3600	11.9%	73.0%	61/355	34/185	60	29	3/5
7	7645	80	120/399	66/256	40	45	5/10	8950	3600	17.1%	89.3%	115/361	91/286	40	39	4/7
8	4877	63	70/315	45/234	37	27	2/7	5373	3600	10.2%	81.4%	86/385	46/268	37	39	7/9
9	3954	53	47/238	42/185	3	20	3/5	4443	3600	12.4%	80.3%	59/276	44/199	3	28	4/5
10	4078	50	55/287	40/227	158	29	3/6	4443	3600	9.0%	78.9%	53/227	47/198	104	27	3/4
Avg.	4346	58	54/249	45/201	60	26.2	3/5.8	4756	3600	8.6%	75.3%	60/264	49/203	94	25.1	3.5/5.1
Max.	7645	80	120/399	66/256	158	45	5/10	8950	3600	17.1%	89.3%	115/385	91/286	238	39	7/9

Table 16
Basic problem case (46 aircraft, 90 operations/hour) with taxi costs.

	Inst.	DP		MIP with CPLEX			
		Obj. value	CPU (s)	Obj. value	CPU (s)	Gap (DP)	Gap (LB)
2 interdependent runways	1	4613	36	4767	3600	3.2%	24.0%
	2	3925	27	3925	3600	0.0%	12.1%
	3	4686	30	5339	3600	12.2%	44.0%
	4	4327	23	4348	3600	0.5%	9.7%
	5	4746	55	4833	3600	1.8%	17.0%
	6	5423	26	5836	3600	7.1%	38.8%
	7	5199	31	6041	3600	13.9%	44.3%
	8	3838	25	3850	3600	0.3%	19.3%
	9	8126	33	8418	3600	3.5%	50.0%
	10	5418	25	5619	3600	3.6%	41.7%
	Avg.	5030	31	5298	3600	4.6%	30.1%
	Max.	8126	55	8418	3600	13.9%	50.0%
2 independent runways	1	4145	37	4157	3600	0.3%	8.7%
	2	3421	29	3421	3600	0.0%	1.5%
	3	4306	32	4480	3600	3.9%	37.4%
	4	3897	25	3897	3600	0.0%	2.7%
	5	4425	75	4539	3600	2.5%	7.3%
	6	4563	28	4563	3600	0.0%	15.5%
	7	4494	35	4747	3600	5.3%	26.5%
	8	3585	26	3585	3600	0.0%	15.3%
	9	6867	38	7397	3600	7.2%	44.0%
	10	5048	26	5207	3600	3.1%	39.7%
	Avg.	4475	35	4599	3600	2.2%	19.9%
	Max.	6867	75	7397	3600	7.2%	44.0%

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